OMB No. 3145-0279 (Expiration Date 10/31/2026)

NIH BIOGRAPHICAL SKETCH COMMON FORM

Name: Matthis, Jonathan Samir

Persistent Identifier (PID) of the Senior/Key Person: https://orcid.org/0000-0003-3683-646X

Position Title: President

Organization and Location: FreeMoCap Foundation, Inc., Boston, Massachusetts, United States

PROFESSIONAL PREPARATION

INSTITUTION AND LOCATION	DEGREE	Start Date	Completion Date	FIELD OF STUDY
University of Texas at Austin, Austin, Texas, United States	Postdoctoral Fellow	09/2014	06/2019	Perceptual Systems
Rensselaer Polytechnic Institute, Troy, New York, United States	Doctor of Philosophy (PHD)	05/2011	05/2014	Cognitive Science
Rensselaer Polytechnic Institute, Troy, New York, United States	Master of Science (MS)	09/2008	05/2011	Cognitive Science
University of Maryland Baltimore County, Baltimore, MD, United States	Bachelor of Arts (BA)	09/2005	05/2007	Philosophy

Appointments and Positions

2021 - present President, FreeMoCap Foundation, Inc., Boston, Massachusetts, United States
2021 - present President/CEO, FreeMoCap Foundation, Inc., Boston, Massachusetts, United States
2021 - present President, FreeMoCap Foundation, Inc. , Boston, Massachusetts, United States
2019 - 2026 Assistant Professor, Northeastern University, Boston, Massachusetts, United States

Products

Products Closely Related to the Proposed Project

1. Matthis JS, Yates JL, Hayhoe MM. Gaze and the Control of Foot Placement When Walking in Natural Terrain. Curr Biol. 2018 Apr 23;28(8):1224-1233.e5. PubMed Central PMCID: PMC5937949.
2. Matthis JS, Barton SL, Fajen BR. The critical phase for visual control of human walking over complex terrain. Proc Natl Acad Sci U S A. 2017 Aug 8;114(32):E6720-E6729. PubMed Central PMCID: PMC5558990.
3. Jonathan Samir Matthis, Philip Queen, Aaron Cherian, Trent Wirth, Endurance Idehen. FreeMoCap: A free, open source markerless motion capture system. 2026 August. Available from: https://zenodo.org/doi/10.5281/zenodo.22131400 DOI: 10.5281/ZENODO.22131400
4. jonmatthis, Endurance Idehen, philipqueen, Mac Prible, Aaron Cherian. SkellyCam: The camera backend for the FreeMoCap software. 2026 April. Available from: https://zenodo.org/doi/10.5281/zenodo.19921107 DOI: 10.5281/ZENODO.19921107
5. Matthis JS, Muller KS, Bonnen KL, Hayhoe MM. Retinal optic flow during natural locomotion. PLoS Comput Biol. 2022 Feb;18(2):e1009575. PubMed Central PMCID: PMC8896712.

Other Significant Products Highlighting Contributions to Science

1. Matthis JS, Barton SL, Fajen BR. The biomechanics of walking shape the use of visual information during locomotion over complex terrain. J Vis. 2015 Mar 18;15(3) PubMed Central PMCID: PMC4365914.
2. Matthis JS, Fajen BR. Visual control of foot placement when walking over complex terrain. J Exp Psychol Hum Percept Perform. 2014 Feb;40(1):106-15. PubMed PMID: 23750964; NIHMSID: NIHMS1050949.
3. philipqueen, jonmatthis, Aaron Cherian. SkellyClicker: For labeling and traing multi-camera video datasets. 2026 April. Available from: https://zenodo.org/doi/10.5281/zenodo.19389889 DOI: 10.5281/ZENODO.19389889
4. philipqueen, jonmatthis. SkellyTracker: The image analysis backend for the FreeMoCap Project. 2023 December. Available from: https://zenodo.org/doi/10.5281/zenodo.10395641 DOI: 10.5281/ZENODO.10395641

5. Aaron Cherian, jonmatthis. freemocap/skellyforge: 2023.12.1005. 2023 December. Available from: https://zenodo.org/doi/10.5281/zenodo.10395649 DOI: 10.5281/ZENODO.10395649

Certification:

I certify that the information provided is current, accurate, and complete. This includes, but is not limited to, information related to current, pending, and other support (both foreign and domestic) as defined in 42 U.S.C. § 6605.

In accordance with Section 10632 of the CHIPS and Science Act of 2022 (42 U.S.C. § 19232), each individual identified as a senior/key person must certify that they are not a party to a malign foreign talent recruitment program.

Research Security Training Requirement for Federal Award Personnel: In accordance with Section 10634 of the CHIPS and Science Act of 2022 (42 U.S.C. § 19234), each individual identified as a senior/key person must certify that they have completed the requisite research security training that meets the requirements specified in Item 2 of Important Notice No. 149 within 12 months prior to proposal submission.

Misrepresentations and/or omissions may be subject to prosecution and liability pursuant to, but not limited to, 18 U.S.C. §§287, 1001, 1031 and 31 U.S.C. §§3729-3733 and 3802.

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NIH BIOGRAPHICAL SKETCH SUPPLEMENT

Name: Matthis, Jonathan Samir

Persistent Identifier (PID) of the Senior/Key Person: https://orcid.org/0000-0003-3683-646X

Position Title: President

Organization and Location: FreeMoCap Foundation, Inc., Boston, Massachusetts, United States

Personal Statement

I am President of the FreeMoCap Foundation and a researcher studying how vision and body mechanics interact to support human movement in natural environments. My work combines experimental studies of perception and action, full-body motion capture, mobile eye tracking, computational analysis, and open-source scientific software development. I earned my Ph.D. in Cognitive Science at Rensselaer Polytechnic Institute, completed postdoctoral training in Perceptual Systems at the University of Texas at Austin, and served as an Assistant Professor at Northeastern University from 2019 to 2026.

My research has examined how people acquire and use visual information to guide walking over complex terrain. Through controlled experiments and measurements of natural behavior, my collaborators and I identified when visual information is needed to guide foot placement, how gaze adapts to terrain, and how coordinated eye and body movements structure the visual signals available during locomotion (Matthis et al., 2017, 2018, 2022). This work has required developing methods that connect measurements of movement to specific questions about sensorimotor control.

I founded the FreeMoCap Foundation to expand access to the tools needed to study human movement. Through the FreeMoCap project, I contribute scientific direction and software development to a free, open-source markerless motion capture system designed to reduce dependence on expensive, specialized equipment. Recent work includes the FreeMoCap system and associated tools for camera acquisition, image analysis, and dataset annotation (FreeMoCap, 2026; SkellyCam, 2026; SkellyClicker, 2026). These efforts connect my experience as an experimental scientist with the practical challenges of building, documenting, and maintaining reusable research software.

Honors

2021 n/a, FreeMoCap Foundation, Inc

Contributions to Science

1.
    1. Establishing how walking biomechanics shape the use of visual information

    Walking over irregular terrain requires visual information about upcoming footholds, but the timing and spatial extent of that information were not well understood. My collaborators and I used controlled manipulations of terrain visibility to investigate how visual information supports accurate stepping. We found that restricting vision to less than approximately two step lengths ahead impaired obstacle avoidance. Subsequent experiments showed that information about a target foothold is particularly important during the latter portion of the preceding step, before the foot leaves the ground.

    Together, these findings support an account in which walkers use vision prospectively to organize the mechanical conditions for an upcoming step. This work connects visual control to the physical dynamics of bipedal locomotion and provides a framework for investigating how sensory limitations affect walking over complex terrain. As first author, I contributed to the experimental methods, execution, analysis, and interpretation of this research with Brett Fajen and Sean Barton (Matthis and Fajen, 2014; Matthis et al., 2015, 2017).

2.
    1. Characterizing the coordination of gaze and foot placement during natural walking

    Laboratory studies provide experimental control but cannot fully reproduce the visual and mechanical demands of natural terrain. To investigate this coordination outside the laboratory, my collaborators and I developed a system combining mobile eye tracking with full-body motion capture during outdoor walking. This approach allowed us to relate gaze to upcoming footholds and the movement of the body across terrains of differing complexity.

We found that walkers adjusted their gaze strategies to the demands of the terrain while maintaining an approximately consistent temporal look-ahead of 1.5 seconds. On more demanding terrain, gaze was closely coupled to future footholds, with patterns reflecting the need to plan subsequent steps. These findings linked visual sampling to the competing demands of stable foot placement and efficient forward movement.
This work provided both a method for studying coordinated perception and action in natural environments and evidence that gaze must be understood in relation to the behavior it supports. As first author, I contributed to development of the measurement approach, data collection, analysis, and interpretation in collaboration with Jacob Yates and Mary Hayhoe (Matthis et al., 2018).

3.

1. Identifying the structure of retinal optic flow during natural locomotion

Theories of visually guided locomotion often rely on simplified descriptions of optic flow that do not capture the combined effects of eye, head, and body movements during walking. My collaborators and I combined measurements of binocular gaze and full-body movement with computational analysis and geometric simulation to characterize the visual motion available during natural locomotion.
We found that head-centered optic flow fluctuates substantially across the gait cycle, whereas fixation on stable environmental locations produces more regular retinal flow organized around the point of gaze. The structure of this retinal flow contains information about the walker's movement relative to the fixated location. These findings motivate evaluating theories of visual locomotor control using the sensory signals generated by natural behavior and identify candidate visual information for guiding movement.
I contributed to the study's conceptualization, methodology, software, analysis, and writing. We also released the underlying data and analysis code to support reproducibility and further investigation (Matthis et al., 2022).

4.

1. Developing accessible, open-source infrastructure for human movement research

The cost and complexity of specialized motion capture equipment can limit participation in human movement research and constrain where measurements can be collected. I initiated the FreeMoCap project to address these barriers and founded the FreeMoCap Foundation to support its continued development.
Working with an interdisciplinary team, I have contributed to FreeMoCap, a free, open-source markerless motion capture system, and an associated collection of reusable software tools. These products include SkellyCam for camera acquisition, SkellyTracker for image analysis, and SkellyClicker for annotation of multi-camera video datasets. Together, these efforts address practical steps needed to turn recorded video into usable movement data.
The scientific contribution is an accessible and inspectable software foundation that researchers can adapt, evaluate, and extend for their own measurement needs. Making these tools available supports methodological transparency and creates opportunities for research and training beyond laboratories with specialized motion capture infrastructure. My role combines project initiation, scientific direction, software contributions, and leadership of the nonprofit organization supporting the work (FreeMoCap, 2026; SkellyCam, 2026; SkellyTracker, 2023; SkellyClicker, 2026).

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